

Probability and Basic Combinatorics

Practice Problem Set

Question 1: The Dice Sequence

Problem Statement

Three fair, six-sided dice are rolled simultaneously. What is the probability that all three dice show completely different numbers?

Question 2: The Seating Arrangement

Problem Statement

Five friends (Alice, Bob, Charlie, Dave, and Eve) sit in a random order in a single row at a movie theater. What is the probability that Alice and Bob sit exactly next to each other?

Question 3: The Even PIN Code

Problem Statement

A 4-digit PIN is formed using the digits 1, 2, 3, and 4 exactly once each. What is the probability that the resulting PIN is an even number?

Question 4: The Quality Control Check

Problem Statement

A shipment contains 10 working electronic components and 3 defective ones. If a technician randomly selects 4 components for testing, what is the probability that all 4 are working perfectly?

What is the probability that exactly 2 of them are working ?

Question 5: The Coin Toss Distribution

Problem Statement

A fair coin is tossed 5 times in a row. What is the probability of observing exactly 3 heads?

What is the probability of seeing 3 consecutive heads?

Question: Full House vs. Flush

Problem Statement

Five cards are drawn at random from a standard, well-shuffled 52-card deck.

Which event has a higher probability of occurring: drawing a Full House or drawing a Flush? Prove your answer by calculating the total number of combinations for each hand.

Recall: A Full House consists of three cards of one rank and two cards of another. A Flush consists of five cards of the same suit.

Question 7: The “At Least One” Defect

Problem Statement

Using the same shipment from Question 4 (10 working components, 3 defective), what is the probability that a random sample of 4 components contains **at least one** defective unit?

Question 8: Overlapping Academic Subjects

Problem Statement

In a class of 40 students, 25 are enrolled in Mathematics, 15 are enrolled in Physics, and 10 are enrolled in both. If a student is chosen at random, what is the probability that they study **neither** Mathematics nor Physics?

Question 9: Mutually Exclusive Card Draws

Problem Statement

Two cards are drawn simultaneously from a standard 52-card deck. What is the probability that the two cards are either both face cards (Jacks, Queens, Kings) or both Aces?

Question 10: The Derangement Problem

Problem Statement

Four employees hand their coats to a cloakroom attendant. At the end of the night, the attendant has lost the tickets and hands the coats back completely at random. What is the probability that **no one** receives their own coat?

What happens if there are n employees and $n \rightarrow \infty$.